

# Seiberg Witten theory Simply.

- Review on Quantum Field theory
  - Seiberg Witten Introduction
- 

## Review on gauge theory.

### ► Classical E & M

Field      Gauge field  
 $E, B \leftrightarrow \phi, A,$

$$m^2_0 = Q(E + v \times B)$$

$$\rightarrow \begin{cases} E = -\nabla\phi - \partial_t A \\ B = \text{rot } A \end{cases}$$

### ► (U(1)) Gauge theory

$$D_\mu = \partial_\mu - iQ A_\mu \quad \text{Covariant derivative}$$

connection

$$[D_\mu, D_\nu] = -iQ F_{\mu\nu} \quad \text{Field tensor.}$$

$$\phi \rightarrow \phi' = e^{-iQ\theta(x)} \phi \quad \text{U(1) Gauge Transformation}$$

$\Rightarrow$  SU(2), SU(3) Non abelian gauge theory.

Standard theory.

•  $SU(3)_C \times \underline{SU(2)_L} \times U(1)_Y$  Gauge theory  
 强相互作用 电弱统一

C: color

L: Left

(Spontaneous symmetry breaking)

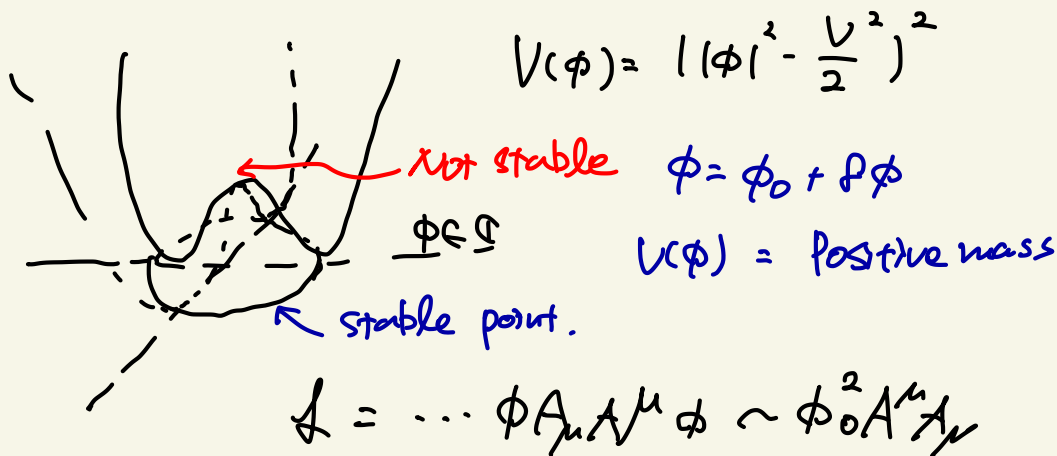
$$SU(2)_L \times U(1)_Y \xrightarrow{SSB} U(1)_{EM}$$

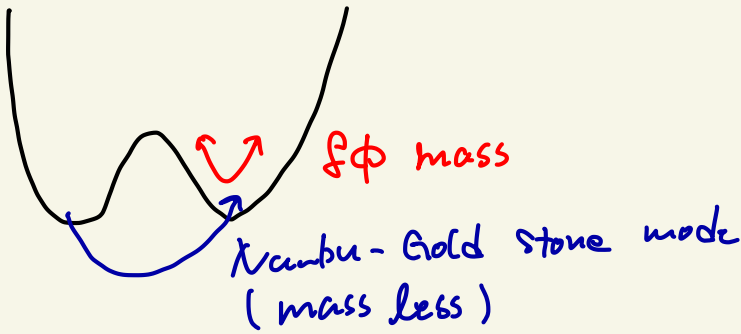
Yang "Weak theory is  $SU(2)$  gauge theory?"

Pauli "Contradiction between gauge invariant and mass term."

$m^2 A_\mu A^\mu$  Not Gauge invariants

→ Acquire mass term through SSB.





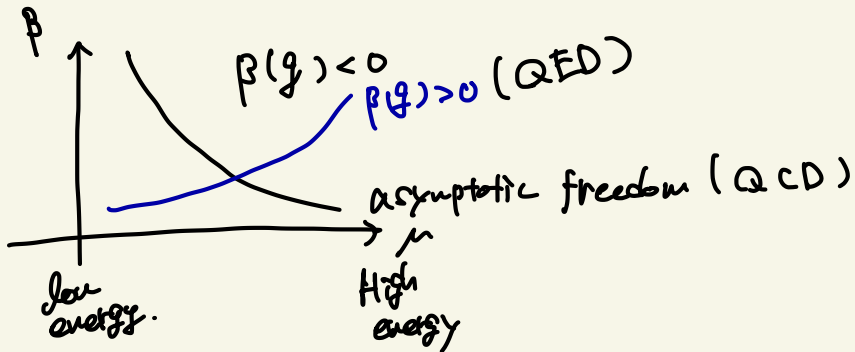
Beta function (SU(N) gauge theory with functional representation)

$$\beta = \mu \frac{dg(\mu)}{d\mu} = - \frac{g^3}{4\pi^2} (11N - 2Nf)$$

$g$ : Coupling Constant.

# of Fermion

$\mu$ : Energy scale



→ Effective theory will be different from different energy scale.

# • QED Spinor 4

Coupling Constant is small at low energy scale

→ We can use perturbation

Perturbation?

Gauss:  $\int dx e^{-\frac{a}{2}x^2} = \sqrt{\frac{2\pi}{a}}$

$$I(\lambda) = \int dx e^{-\frac{a}{2}x^2 - \lambda x^4} \quad (\lambda > 0)$$

$$= \int dx e^{-\frac{a}{2}x^2} \left( \sum_{n=0}^{\infty} \frac{(-\lambda)^n}{n!} x^{4n} \right)$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{1}{n!} (-\lambda)^n \int_{-\infty}^{\infty} dx e^{-\frac{a}{2}x^2} x^{4n}$$

Exchange, but  $\otimes$  can not be changed and moreover it is not even converging when  $\lambda > 0$ .  
So is this wrong?

$$\sum_{n=0}^{\infty} \frac{1}{n!} (-\lambda)^n \int_{-\infty}^{\infty} e^{-\frac{a}{2}x^2} x^{4n}$$

→  $\lambda \ll 1$ , only see low order of  $\lambda$ .

Perturbation

(Increasing  $n$  will be different from true value, ex)  $d \sim (37, 137)$  order.

Assumption we made is  $\lambda$  to be small.  
Only then perturbation is possible.

$\lambda \gg 1$  (Strong Coupling).

In QCD, it is strong coupling in low energy scale. In theory it is possible to use renormalization from high energy QCD.  
But it is difficult.

(QCD) How do we analyze?

Seiberg-dual theory will give general theory of 4d  $N=2$  SU(2) Super Yang Mills theory without any approximation.

Use of duality.

What is duality?

There is some connection between  
different theory.

ex) Bramers-Hannier duality

Maxwell ( $\rho=0, \dot{\rho}=0$ )

$$(\mathbb{E}, \mathbb{B}) \rightarrow (\mathbb{B}, -\mathbb{E})$$